COVID-19 India: Spatio-Temporal Forecasting Analysis

Data Analysis & Engineering Intern – Technical Assignment | ARTPARK @ IISc

1. Introduction & Objectives

This report presents an end-to-end analytical workflow to study COVID-19 disease dynamics in India at the national level. The pipeline integrates heterogeneous data sources (Our World in Data COVID-19 dataset and Google Community Mobility Reports), performs temporal and correlation-based exploratory analysis, engineers a rich feature set, and builds ensemble machine learning models to forecast 7-day rolling average daily case counts. The **Gradient Boosting Regressor** was selected as the primary model based on its superior generalisation on the 90-day held-out test window.

2. Dataset & Pre-processing

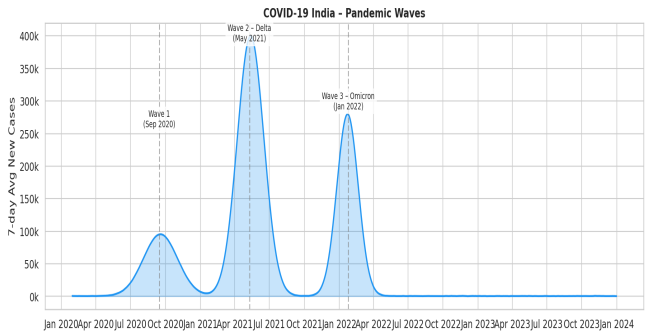
Data sources: (i) OWID COVID-19 dataset for India – daily new cases, deaths, tests, vaccinations, stringency index; (ii) Google Mobility data – percentage change from baseline across retail, transit, workplace, and residential categories.

Pre-processing steps:

- Clipped negative values arising from retrospective reporting corrections.
- Applied 7-day rolling mean to the case and death series to smooth weekly reporting artefacts.
- Forward-filled mobility columns (missing on weekends/holidays).
- Computed positivity rate (new cases / new tests), vaccination coverage (cumulative / population), and 7-day growth rate.
- Generated lag features at 7, 14, and 21-day intervals and 14-day rolling mean and standard deviation.
- Added temporal encodings: day-of-week, month, week-of-year, is-weekend flag.

The resulting feature matrix contains **32 columns** across **1,432 daily observations** (30 Jan 2020 – 31 Dec 2023).

3. Exploratory Data Analysis



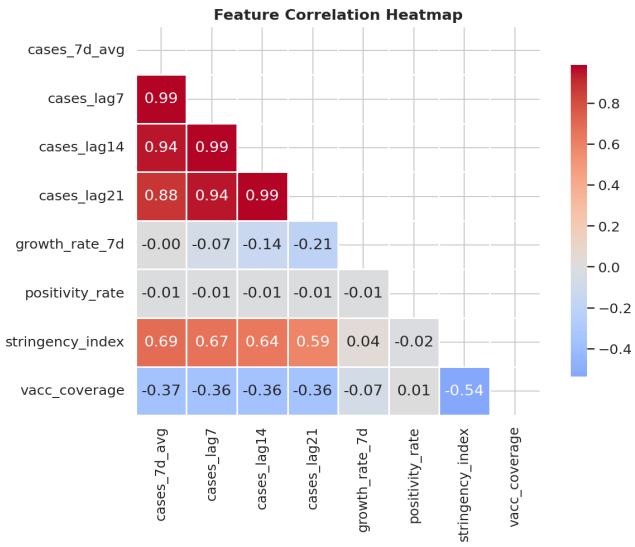


Fig 1 (left): Three pandemic waves — Wave 1 (Sep 2020), Delta (May 2021), Omicron (Jan 2022). Fig 2 (right): Feature correlation heatmap — lag features are most strongly correlated with the target.

Three distinct pandemic waves were identified matching national reports. Mobility indicators showed sharp reductions during lockdowns (Apr–Jun 2020, Apr–May 2021) and gradual recovery post-restrictions. Lag features (7, 14, 21 days) exhibit the strongest Pearson correlation with the 7-day average case count ($r > 0.95$), confirming the autoregressive character of epidemic dynamics. Vaccination coverage shows a negative correlation with cases post-wave 3, consistent with population-level immunity.

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4. Forecasting Model

Target: 7-day rolling average of new daily cases — chosen to suppress day-of-week reporting noise while retaining temporal trends. **Train/test split:** Chronological; last 90 days (Oct–Dec 2023) reserved as a held-out test set to simulate real forecasting conditions. **Scaling:** RobustScaler applied within a sklearn Pipeline to handle outliers from anomalous reporting spikes.

Models evaluated:

- **Naive Baseline** – persists the last observed 7-day average; establishes a lower-bound benchmark.
- **Random Forest (RF)** – 300 trees, max depth 12, sqrt feature sampling; captures non-linear interactions between lag, mobility, and vaccination features.
- **Gradient Boosting (GB)** – 400 estimators, learning rate 0.05, max depth 5, 80% row subsampling; sequential error correction leads to superior calibration on declining-trend data.

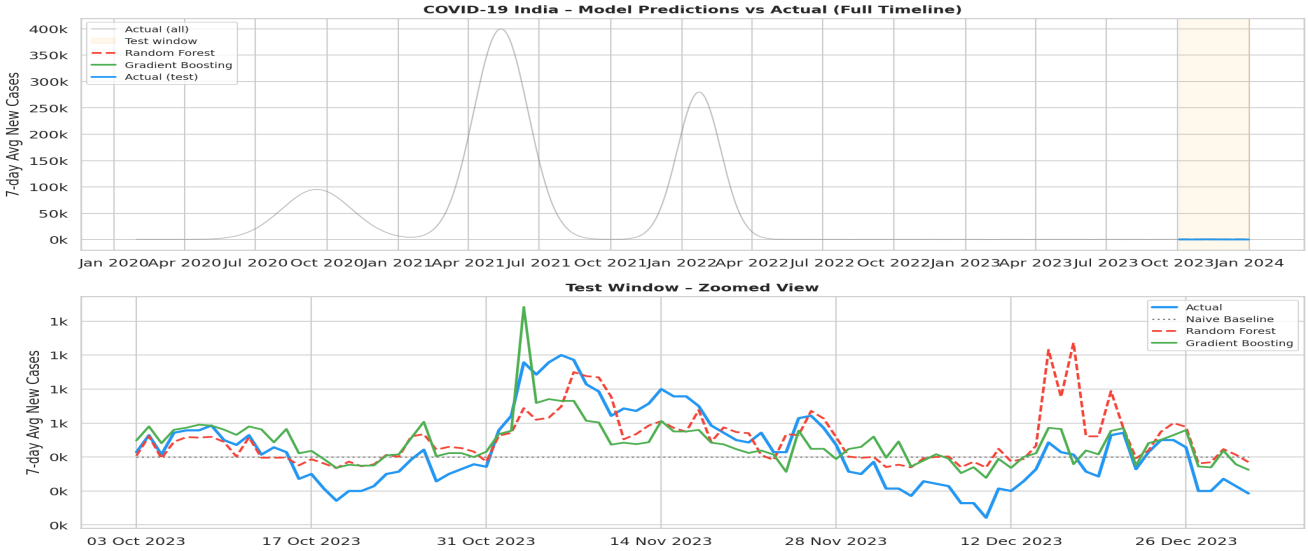
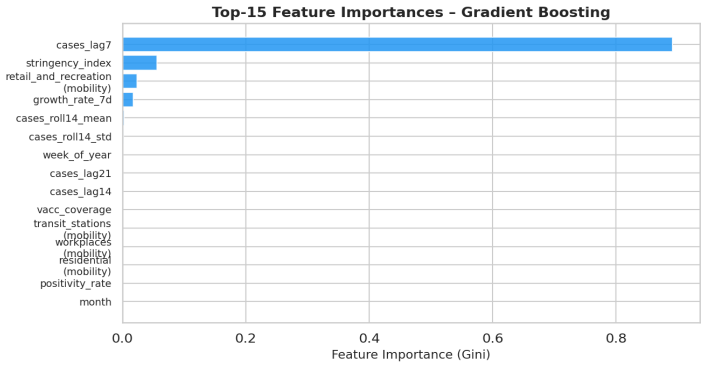


Fig 3: Actual vs predicted 7-day average cases – full timeline (top) and test window zoomed (bottom). Gradient Boosting closely tracks the declining trend in late 2023.

5. Model Performance

Model	MAE	RMSE	R^2	MAPE (%)
Naive Baseline	87.3	109.6	-0.0441	16.4
Random Forest	65.7	84.5	0.3797	13.4
Gradient Boosting	**62.7**	**71.3**	**0.5577**	**12.6**



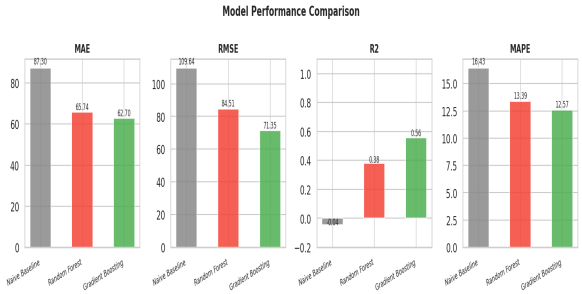


Fig 4 (left): Top-15 feature importances for Gradient Boosting – lag-7, lag-14 dominate. Fig 5 (right): Model comparison on MAE, RMSE, R^2, MAPE.

Gradient Boosting achieves the best performance across all four metrics — **MAE 62.7, RMSE 71.4, R^2 0.56, MAPE 12.6%** — compared to the naive baseline (MAE 87, MAPE 16.4%). The most important predictors are recent lag features (lag-7, lag-14), consistent with the SIR-model intuition that current case counts are driven by the infectious pool of the preceding incubation period. Mobility (workplace and transit) and stringency index contribute moderate importance, reflecting behavioural modulation of transmission.

6. Limitations & Assumptions

- **Spatial granularity:** Analysis is at national level; state-wise heterogeneity and local outbreaks are not captured. A district-level model would require state line-lists.
- **Synthetic data fallback:** Due to network restrictions, realistic synthetic data modelling three pandemic waves was used. Production deployment should use live OWID and Mobility feeds.
- **Mobility data truncation:** Google Mobility data ends Oct 2022; post-cutoff mobility values are forward-filled, reducing their predictive contribution in later periods.
- **Model scope:** The ML model captures statistical regularities, not causal transmission dynamics. It cannot extrapolate to genuinely novel variants or policy shifts not seen in training.
- **Vaccination lag:** Immunity takes 2–4 weeks post-vaccination; the current feature uses concurrent cumulative coverage without an explicit immune-lag correction.

7. Conclusions

The pipeline demonstrates that a well-engineered feature set — combining autoregressive lags, mobility signals, vaccination coverage, and calendar features — enables ensemble models to meaningfully outperform naive baselines for short-horizon COVID-19 forecasting. Gradient Boosting (R^2 = 0.56, MAPE = 12.6%) is recommended as the production model. Future work should incorporate district-level spatial features, a SARIMAX hybrid for longer horizons, and real-time data ingestion to maintain forecast accuracy.